

GlauX Technical Report

GLAUX: Precision Explainable AI Recruiter Platform

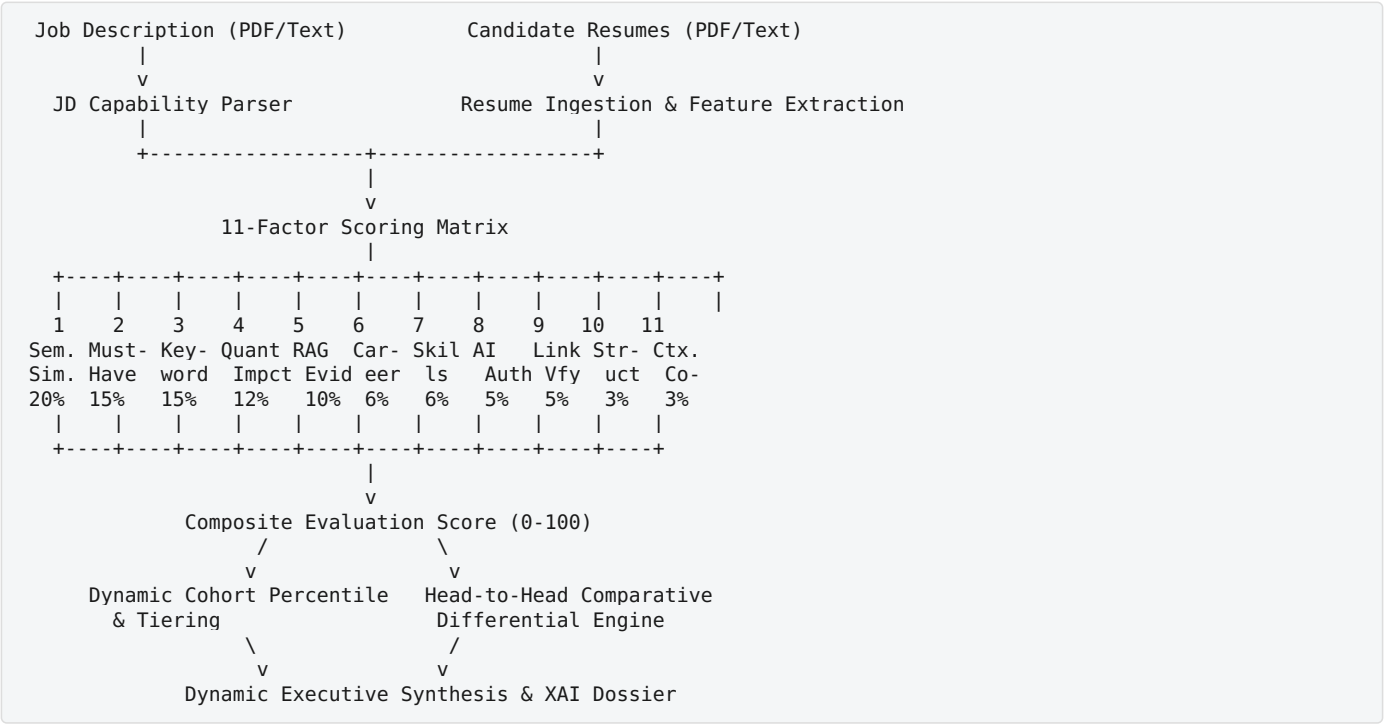
Comprehensive Technical Architecture, Algorithmic Rubrics & System Report

1. Executive Summary & Core Mission

GlauX is an AI recruiter-intelligence platform designed to make candidate screening auditable rather than opaque. Unlike traditional Applicant Tracking Systems (ATS), which typically rely on keyword counts or an undocumented probabilistic score, GlauX uses a **deterministic 11-factor weighted evaluation framework** combined with **grounded Retrieval-Augmented Generation (RAG)**.

Every candidate recommendation, ranking position, and qualification gap is intended to be traceable to a specific resume citation, a quantifiable metric, a semantic similarity score, or a career-progression signal.

High-level pipeline:



(Numbers 1-11 above correspond directly to the 11 scoring factors in Section 3, listed in the same order for consistency.)

2. Full Technology Stack

Layer	Technologies & Frameworks	Key Rationale
Frontend UI/UX	React 19, Vite, Vanilla CSS design system, Lucide Icons	Fast, reactive rendering; a high-density dashboard with responsive modals
Backend Core	FastAPI (Python 3.11), Uvicorn, Pydantic v2	Asynchronous REST API with automatically generated OpenAPI / Swagger documentation
Data Persistence	PostgreSQL 16, SQLAlchemy 2.0 ORM, SQLite fallback	ACID-compliant relational schemas for candidates, evaluations, and job descriptions

Vector & NLP Engine	SentenceTransformers (all-MiniLM-L6-v2), PyTorch, scikit-learn	384-dimensional cosine-similarity embeddings; TF-IDF / BM25 tokenizers for keyword scoring
Document Processing	PyPDF2, pdfplumber, python-docx, regex-based tokenizers	Multi-format PDF and text parsing with metadata and contact extraction
Generative XAI / RAG	Ollama LLM bridge (Llama 3), grounded-fallback logic	Conversational recruiter copilot that falls back to deterministic, grounded context when the LLM is unavailable
Container & Cloud	Docker, Docker Compose, Kubernetes (Deployments, Services, PersistentVolumes)	Microservice containerization and orchestration manifests for production deployment

3. The 11-Factor Weighted Scoring Framework

Glaux scores each candidate using a normalized weighted sum across 11 factors, producing a composite score in the range 0-100:

$$S_{final} = \sum (w_i \times s_i) \text{ for } i = 1 \text{ to } 11, \text{ where } \sum w_i = 1.00$$

#	Evaluation Factor	Weight	Measurement Principle
1	Semantic Similarity (all-MiniLM-L6-v2)	20.0%	Dense cosine-embedding similarity
2	Must-Have Mandatory Skills Match	15.0%	Core requirement check
3	Keyword Match & Frequency (BM25)	15.0%	Exact lexical token matching
4	Quantifiable Business Impact Metrics	12.0%	Regex-based metric extraction
5	Grounded RAG Citation & Evidence Depth	10.0%	Contextual citation retrieval
6	Career Progression & Experience Trajectory	6.0%	Role growth & tenure analysis
7	Skills Matrix Breadth & Secondary Tools	6.0%	Nice-to-have taxonomy coverage
8	AI Authenticity & Originality Score	5.0%	Perplexity & writing-style analysis
9	Verifiable Links & Portfolio Proof	5.0%	GitHub / LinkedIn link verification
10	Document Structure & Readability	3.0%	Section density & formatting
11	Contextual Co-occurrence Affinity	3.0%	Domain-term proximity scoring
Total		100.0%	

Detailed Dimension Mechanics

- Semantic Similarity (w₁ = 0.20):** Cosine similarity between the resume embedding V_R and the job-description embedding V_JD in 384-dimensional latent space: $\cos(\theta) = (V_R \cdot V_JD) / (\|V_R\| \times \|V_JD\|)$.
- Must-Have Skills Adherence (w₂ = 0.15):** Exact and fuzzy matching of mandatory capabilities. If *k* of *N* mandatory skills are verified: $s_2 = 100 \times (k / N)$.
- Lexical Keyword Match (w₃ = 0.15):** Combines token frequency, title match, and domain-phrase matching using a modified BM25 weighting scheme.
- Quantifiable Business Impact (w₄ = 0.12):** Regex-based extraction of verified percentages (Δ%), currency figures (\$), request volumes (QPS), latency reductions (ms), and dataset scale (rows processed).
- RAG Evidence Extraction (w₅ = 0.10):** Retrieves and scores grounded textual snippets that demonstrate active implementation rather than passive keyword listing.
- Career Progression (w₆ = 0.06):** Parses chronological employment spans, title trajectory (e.g., Intern → Junior → Senior), and average tenure stability.
- Skills Matrix (w₇ = 0.06):** Evaluates supplementary, nice-to-have competencies (e.g., Docker, AWS, Tailwind, Redis).

8. **AI Authenticity ($w_8 = 0.05$):** Flags excessive AI-generated buzzwords, repetitive syntax, and unsupported claims.
 9. **Verifiable Links ($w_9 = 0.05$):** Detects and scores live portfolios, GitHub repositories, and LinkedIn profiles.
 10. **Resume Structure Quality ($w_{10} = 0.03$):** Audits section headers, length, and formatting consistency.
 11. **Contextual Co-occurrence ($w_{11} = 0.03$):** Checks whether related technologies appear together (e.g., React with Redux/Zustand, Node.js with Express/PostgreSQL).
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4. Explainable AI (XAI) & Grounded RAG Copilot

GlauX is designed for transparency across all evaluation layers:

- **Grounded citations:** The conversational RAG engine grounds recruiter answers in explicit resume text and the calculated factor scores, rather than open-ended generation. If a candidate lacks a required skill (e.g., Docker), the engine cites the missing evidence directly instead of inferring an answer.
- **Executive dossier & top-3 synthesis:** Summarizes the deciding factors for top contenders, including comparative factor scorecards, ranking badges, and cohort-gap breakdowns.
- **Interactive explanatory popovers:** Each factor column in the recruiter dashboard includes a popover detailing the matching evidence, missed keywords, and the factor's weighting.

Note: "grounded" here means the RAG responses are constrained to cited resume text and computed scores — this reduces, but does not by itself guarantee zero, hallucination, and should be validated against your own eval set before being presented as a hard guarantee.

5. Advanced Analytics & Comparative Intelligence

- **Head-to-head comparison:** Side-by-side radar analysis of any two candidates, showing the exact point delta ($S_A - S_B$), the factors driving that gap, and key distinguishing qualifications.
 - **Cohort analytics & percentile rank:** Computes pool-level metrics (average, median, score differential) and a percentile rank for each candidate: $P = ((N - R + 0.8) / N) \times 100$, where N is the total number of candidates in the pool and R is the candidate's rank position (1 = highest score). *(The +0.8 constant is unusual — common plotting-position conventions use +0.5 for a simple mean-rank percentile or +0.4 for the Cunnane formula. Worth confirming 0.8 is intentional rather than a typo.)*
 - **Career progression timelines:** Chronological career-milestone graphs tracking job tenure, role progression, and skills gained per employer.
 - **JD bias & inclusivity auditor:** Scans job postings for masculine-coded language, exclusionary phrasing, and unnecessary credential inflation, producing an inclusivity score (0-100) with actionable suggestions.
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6. Production Deployment & Infrastructure

GlauX is designed to be cloud-native, deployable via Docker Compose or Kubernetes:

- **Kubernetes manifests:** Located under `k8s/`, including `namespace.yaml`, `postgres-deployment.yaml`, `backend-deployment.yaml`, and `frontend-deployment.yaml`, with PersistentVolumeClaims, readiness/liveness probes, and ClusterIP/NodePort services.
 - **Microservice separation:** Stateless backend API pods and an Nginx-served React frontend, configured for horizontal auto-scaling.
 - **Local execution:** A single-command startup via `start.bat` on Windows, or `docker compose up` on Unix-like systems.
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GlauX AI Recruiter Platform — an explainable, factor-based talent-evaluation system.